

- 8 (a) Fig. 8.1 shows a transmission electron micrograph of a mitochondrion.



Fig. 8.1

- (i) On Fig. 8.1, use the letter **A** with a label line to show a location where the Krebs cycle occurs. [1]
- (ii) Name the structure labelled **B** on Fig. 8.1 that forms part of the inner mitochondrial membrane.

..... [1]

(iii) The inner mitochondrial membrane is the site of oxidative phosphorylation.

Explain how the structure of the inner mitochondrial membrane is linked to its function.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

(b) The activity of succinate dehydrogenase, an enzyme in the Krebs cycle, is used as a measure of the rate of respiration in the mitochondria. The redox reaction catalysed by succinate dehydrogenase is shown in Fig. 8.2.

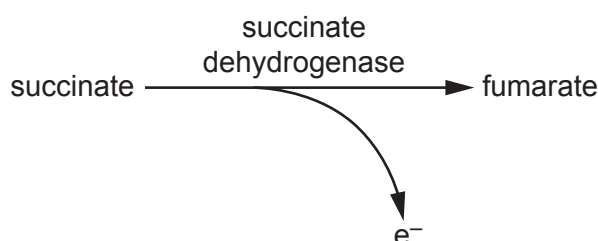


Fig. 8.2

The activity of succinate dehydrogenase can be measured using a redox indicator.

Name a redox indicator **and** explain why this indicator can be used to measure the activity of succinate dehydrogenase.

.....

.....

.....

.....

..... [2]

(c) Fig. 8.3 shows a ground squirrel, *Ictidomys tridecemlineatus*.



Fig. 8.3

In the winter, the ground squirrel curls into a spherical shape underground and sleeps for long periods. During this time, the ground squirrel switches between two states:

- torpor, when the body temperature is maintained at 10 °C
- euthermia, when the body temperature is maintained at 37 °C.

Scientists used the activity of succinate dehydrogenase to investigate the rate of respiration in the mitochondria of ground squirrels. Mitochondria were extracted from liver and muscle samples of ground squirrels. The rate of respiration was measured at different concentrations of succinate and at temperatures that corresponded to torpor (10 °C) and euthermia (37 °C).

The results are shown in Fig. 8.4.

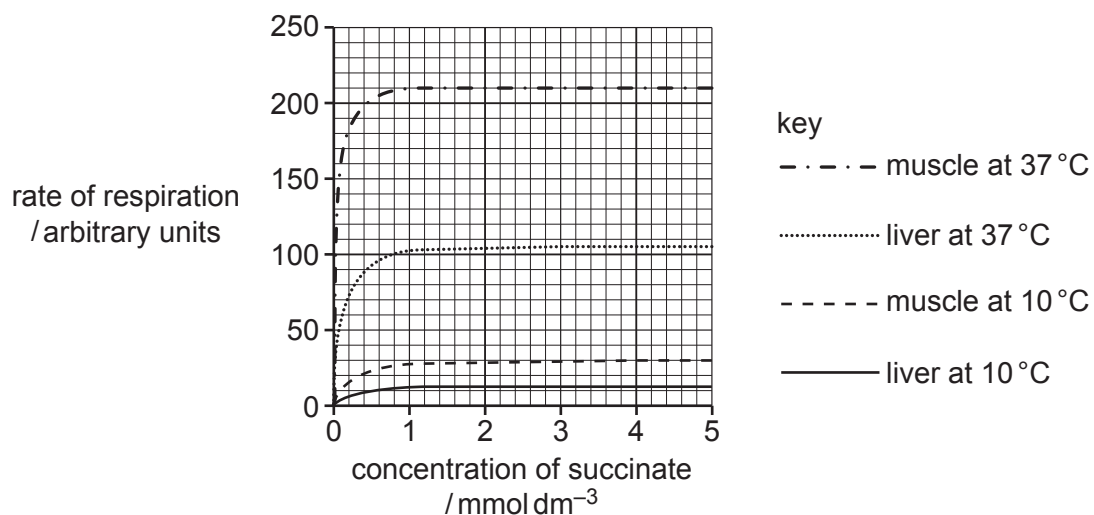


Fig. 8.4

(i) Describe the trends shown in Fig. 8.4.

.....

.....

.....

.....

.....

.....

..... [3]

(ii) Explain the difference in the rates of respiration between liver at 37 °C and muscle at 37 °C.

.....

.....

.....

.....

.....

.....

..... [3]

(iii) During torpor in ground squirrels, muscle tissue uses more energy than liver tissue.
Suggest **one** reason for this difference.

.....

..... [1]

[Total: 16]

Section B

Answer **one** question.